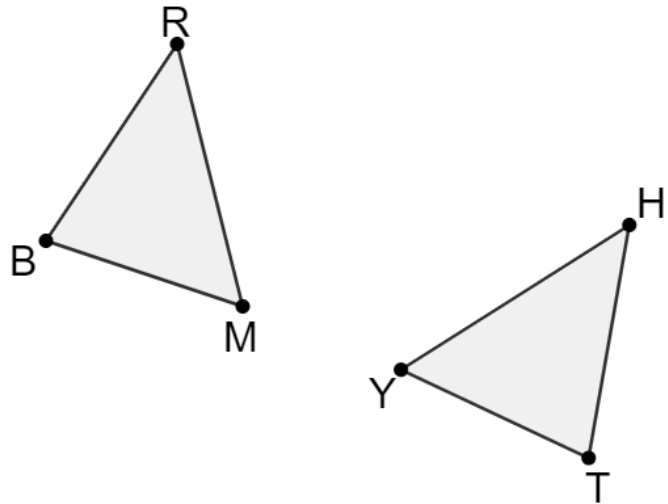


Triangles  $BRM$  and  $YHT$  are shown. Determine if the given information is sufficient to conclude that the triangles are congruent. If so, name the congruence theorem or postulate that shows congruence.



| Given Information                                                                                                  | Are They Congruent? | Theorem or Postulate Used |
|--------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------|
| 1. $\overline{MR} \cong \overline{YH}$ , $\angle HYT \cong \angle BMR$ , $\angle BRM \cong \angle THY$             |                     |                           |
| 2. $\overline{MB} \cong \overline{YT}$ , $\overline{MR} \cong \overline{HY}$ , $\overline{BR} \cong \overline{TH}$ |                     |                           |
| 3. $\overline{MB} \cong \overline{YT}$ , $\angle YTH \cong \angle MBR$ , $\angle BRM \cong \angle THY$             |                     |                           |

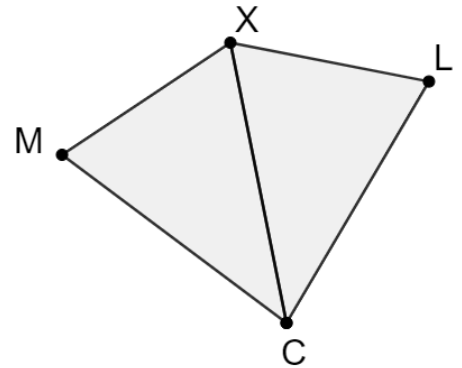
4. Quadrilateral  $CLXM$  is shown where  $\overline{MC} \cong \overline{LC}$  and  $\angle MCX \cong \angle XCL$ .  
Complete the paragraph proof.

It is given that  $\overline{MC} \cong \overline{LC}$  and  $\angle MCX \cong \angle XCL$ .

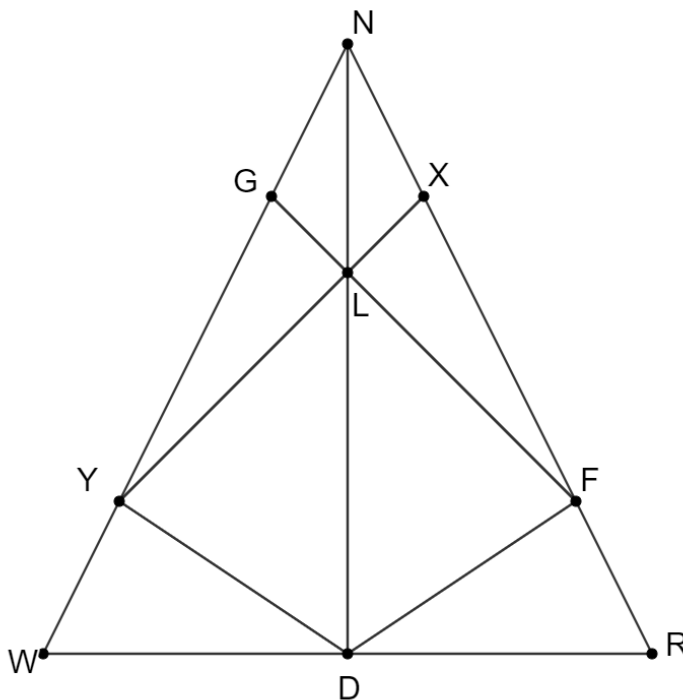
By the reflexive property, \_\_\_\_\_.

Therefore,  $\Delta$ \_\_\_\_\_  $\cong \Delta$ \_\_\_\_\_

by \_\_\_\_\_.



5. Triangle  $NRW$  is shown.

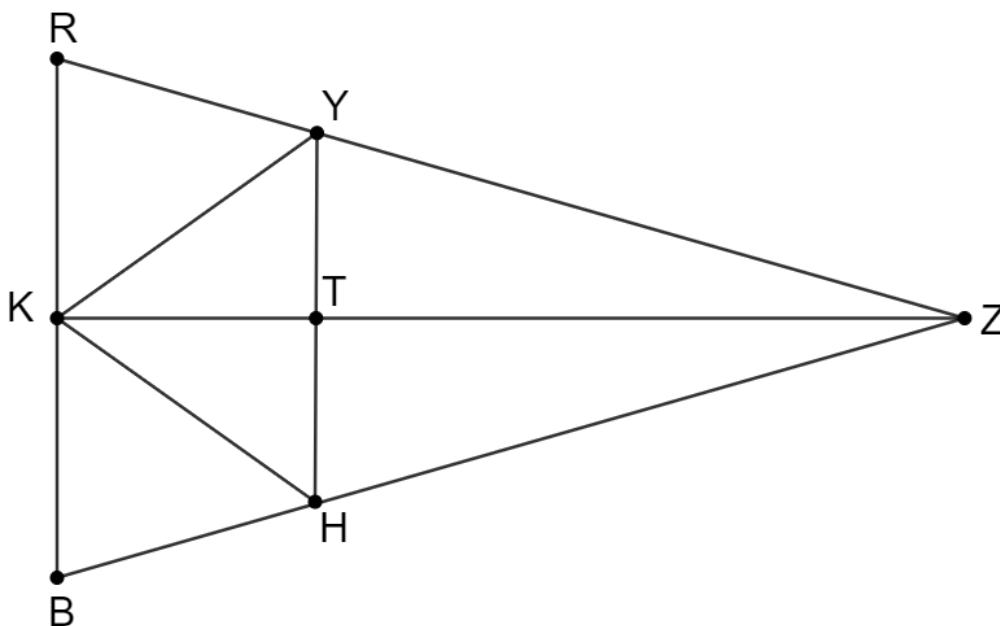


Complete each statement.

- a. If  $\triangle FLD \cong \triangle YLD$  then  $\angle DFL \cong \angle$ \_\_\_\_\_.

- b. If  $\overline{ND} \perp \overline{WR}$  and  $\overline{ND} \cong \overline{ND}$ , the information needed to show that  
 $\triangle NDW \cong \triangle NDR$  using HL is \_\_\_\_\_.

6. Triangle  $RZB$  is shown.



Given:  $\triangle KYT \cong \triangle KHT$

Prove:  $\triangle YKZ \cong \triangle HKZ$

An incomplete proof is shown.

| Statement                              | Reason                                                              |
|----------------------------------------|---------------------------------------------------------------------|
| 1. $\triangle KYT \cong \triangle KHT$ | 1. Given                                                            |
| 2. $\overline{KY} \cong \overline{KH}$ | 2.                                                                  |
| 3.                                     | 3. Corresponding Parts of Congruent Triangles are Congruent (CPCTC) |
| 4.                                     | 4. Reflexive Property                                               |
| 5. $\triangle YKZ \cong \triangle HKZ$ | 5.                                                                  |

What is the reason for step 5?

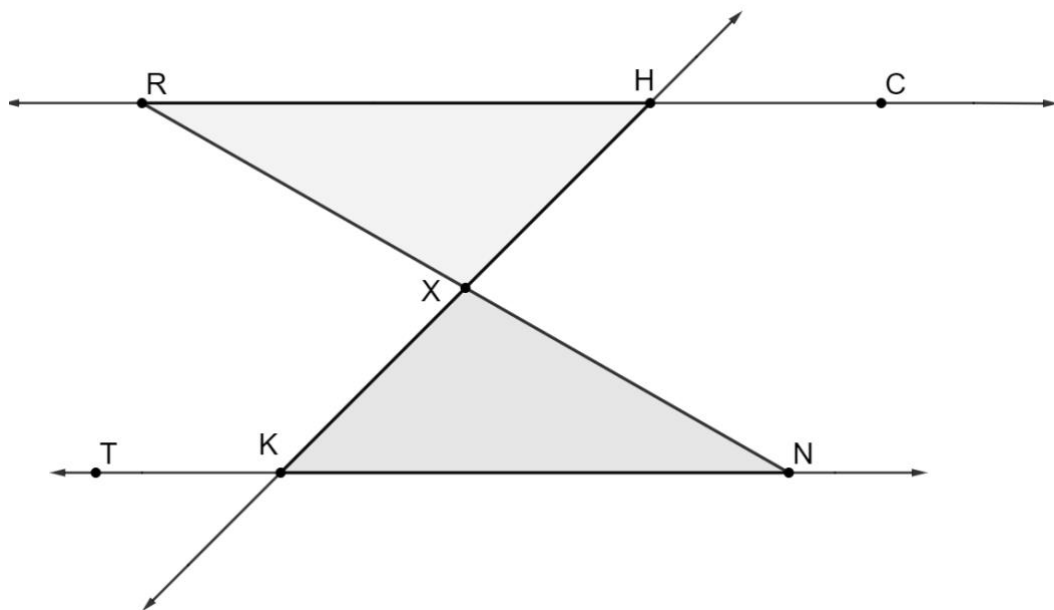
|                          |  |
|--------------------------|--|
| <b>Reason for step 5</b> |  |
|--------------------------|--|

7. Parallel lines  $\overleftrightarrow{RC}$  and  $\overleftrightarrow{TN}$  with transversal  $\overleftrightarrow{KH}$  is shown.

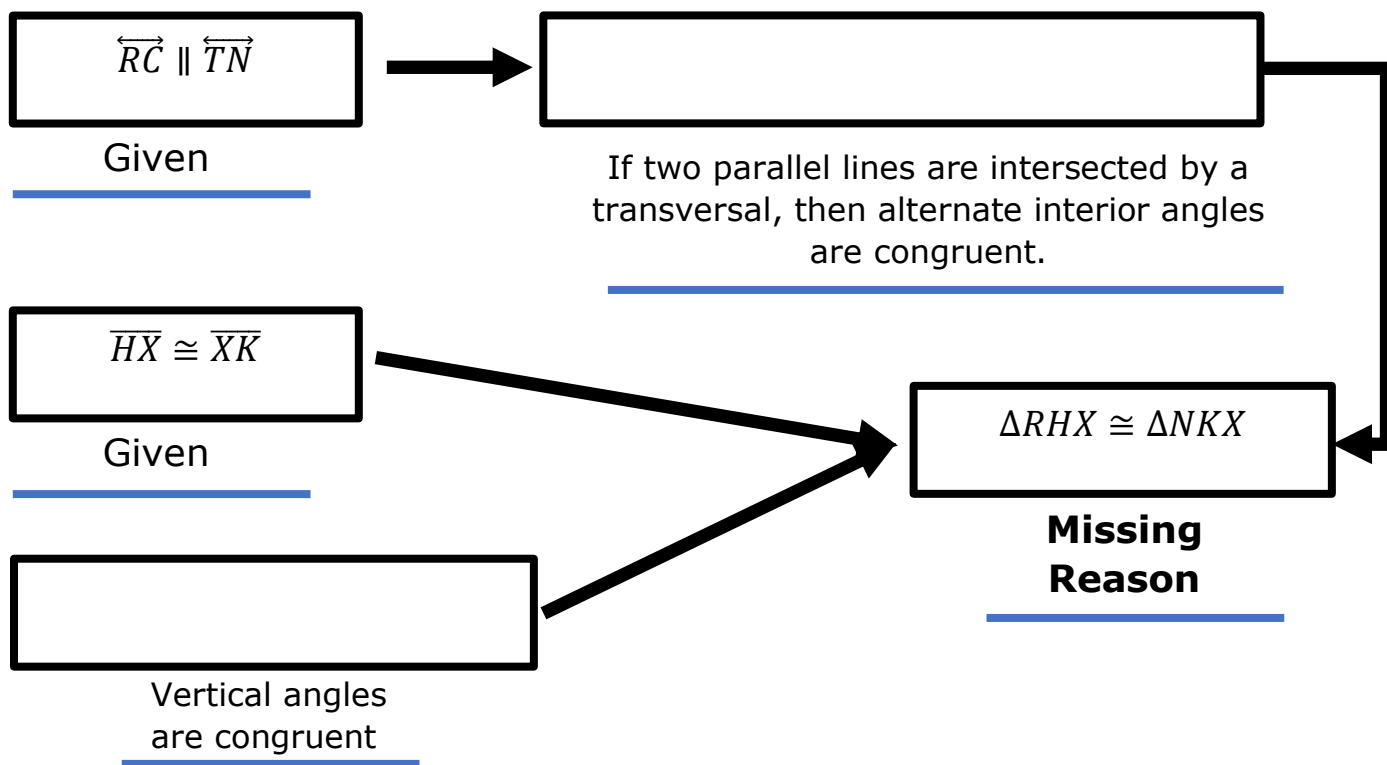
Given:  $\overleftrightarrow{RC} \parallel \overleftrightarrow{TN}$ ,

$\overline{HX} \cong \overline{KX}$

Prove:  $\triangle RHX \cong \triangle NKX$



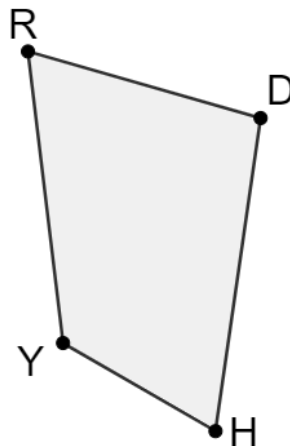
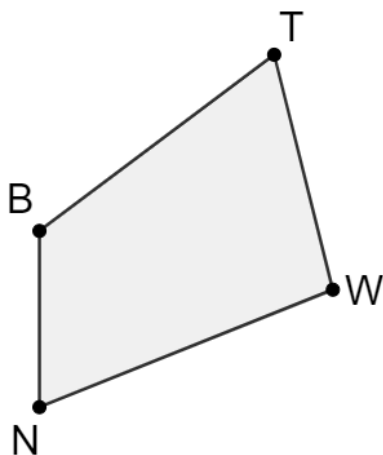
An incomplete flowchart proof is shown.



What is the missing reason in the flowchart proof?

|                       |  |
|-----------------------|--|
| <b>Missing Reason</b> |  |
|-----------------------|--|

8. Quadrilaterals  $TWNB$  and  $HYRD$  are shown.



It is given that  $TWNB \cong RDHY$ ,  $NW = 3x$ ,  $RD = 5x - 19.6$ ,  $RY = 2x + 5.6$ ,  $HD = 5x - 14.4$ ,  $TW = 3x - 5.2$ , and  $BN = x + 4.8$ .

- a. Determine the value of  $x$ . Show your work.

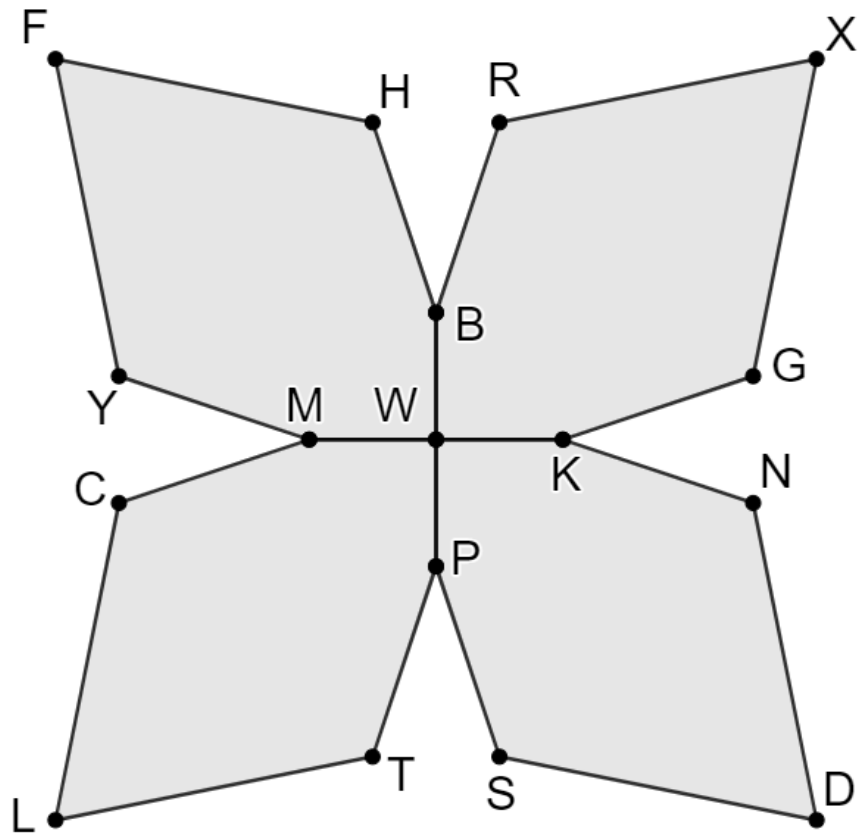
|                                |
|--------------------------------|
| $x = \underline{\hspace{2cm}}$ |
|--------------------------------|

- b. Determine the perimeter of quadrilateral  $TWNB$ . Show your work.

|                  |  |
|------------------|--|
| <b>Perimeter</b> |  |
|------------------|--|

9. A graphic artist created a design for a flower using four congruent hexagons as shown.

Given:  $\overline{BP} \perp \overline{MK}$ ,  
 $m\angle MYF = m\angle FHB$ ,  
 $m\angle WMC = m\angle WPT$ ,  
 $m\angle GXR = 66^\circ$ ,  
 $m\angle DNK = (3x + 18)^\circ$ , and  
 $m\angle WBR = (5x - 8)^\circ$ . The sum of the angles in  $WKNDSP$  is  $720^\circ$ .



Find each. Show your work

a.  $m\angle TLC =$  \_\_\_\_\_

b.  $m\angle KGX =$  \_\_\_\_\_

Work